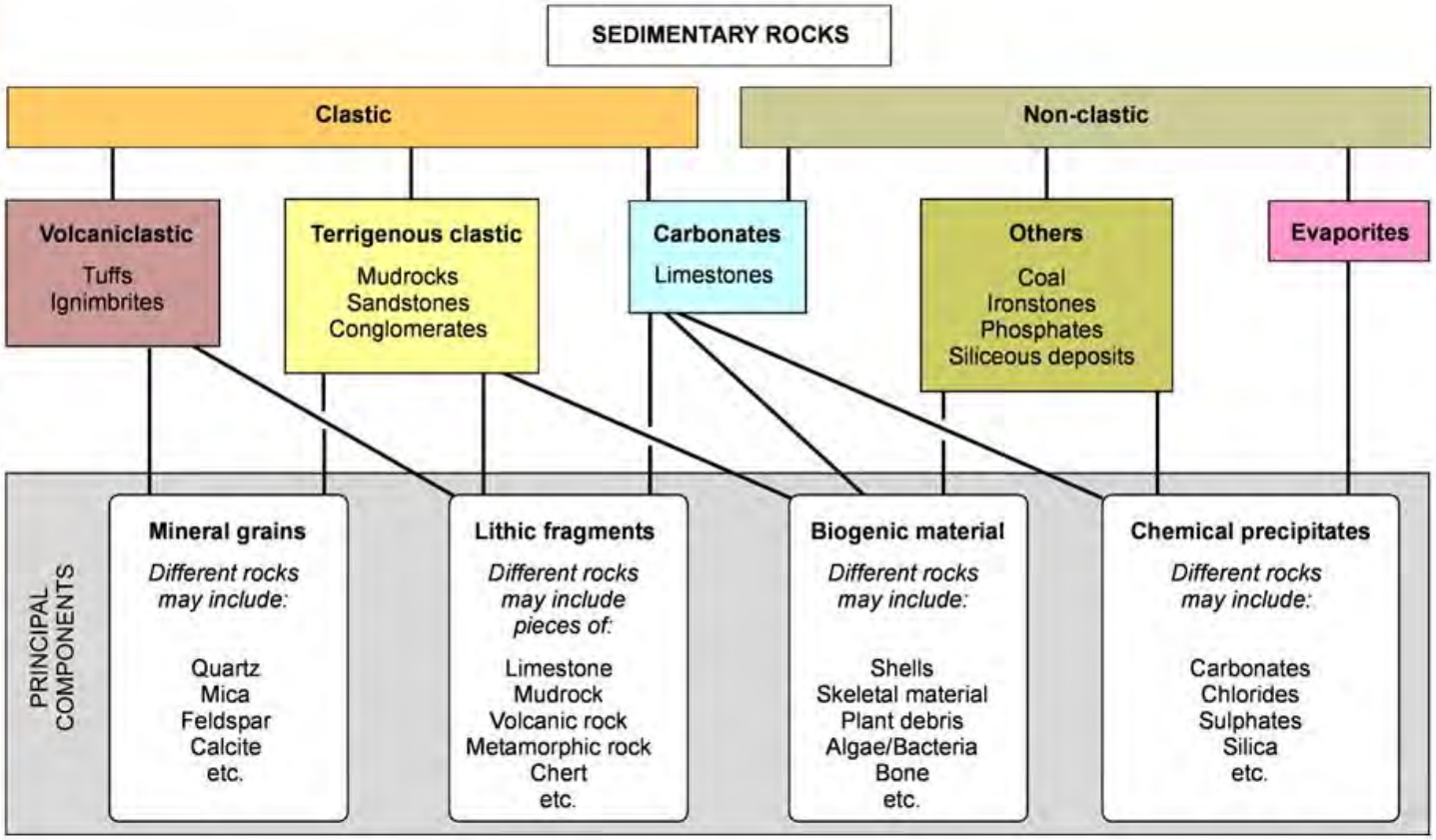
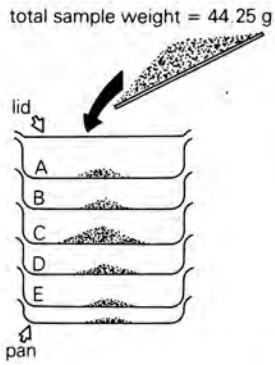
Classification scheme for sediments & sedimentary rocks



Length (mm)	ϕ	Wentworth size class		Sediment/ rock name
			block	mega-conglomerate
4096	-12	vc	boulder	gravel conglomerate
2048	-11	c		
1024	-10	m		
512	-9	f		
256	-8	c	cobble	
128	-7	f		
64	-6	vc	pebble	
32	-5	c		
16	-4	m		
8	-3	f		
4	-2		granule	
2	-1	vc	sand	sand sandstone
1	0	c		
0.50	1	m		
0.25	2	f		
0.125	3	vf		
0.063	4	c	silt	silt siltstone
0.031	5	m		
0.015	6	f		
0.008	7	vf		
0.004	8		clay	clay claystone

Grain size scale for sediments, showing Wentworth size classes, equivalent phi (ϕ) units, corresponding to various millimeter sizes.



Sieve	Screen size (mm)	Weight of sieve (g)
A	1.0	5.31
B	0.5	8.70
C	0.25	12.59
D	0.125	9.13
E	0.0625	6.03
pan	<0.0625	2.49

Figure 7.3 An example of data collected by sieving.

Class (mm)	Class (ϕ)	Weight (g)	Weight (%)
1–2	−1–0	5.31	12.1
0.5–1	0–1	8.70	19.6
0.25–0.5	1–2	12.59	28.5
0.125–0.25	2–3	9.13	20.6
0.0625–0.125	3–4	6.03	13.6
0.0312–0.0625	4–5	2.49	5.6

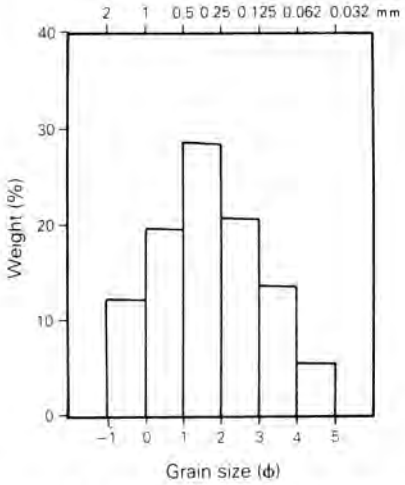


Figure 7.7 A histogram based on data from Fig. 7.3.

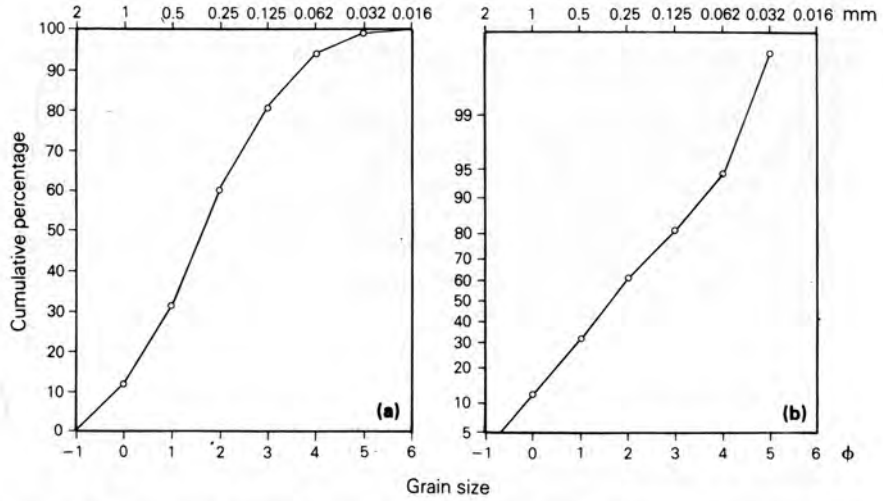
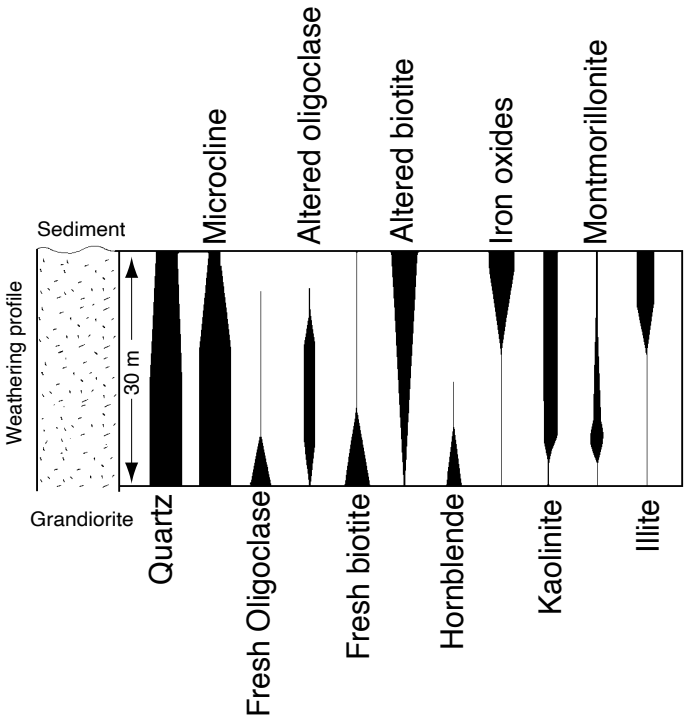


Figure 7.9 Cumulative curves: (a) arithmetic ordinate scale; (b) probability ordinate scale. (Based on data from Fig. 7.3.)

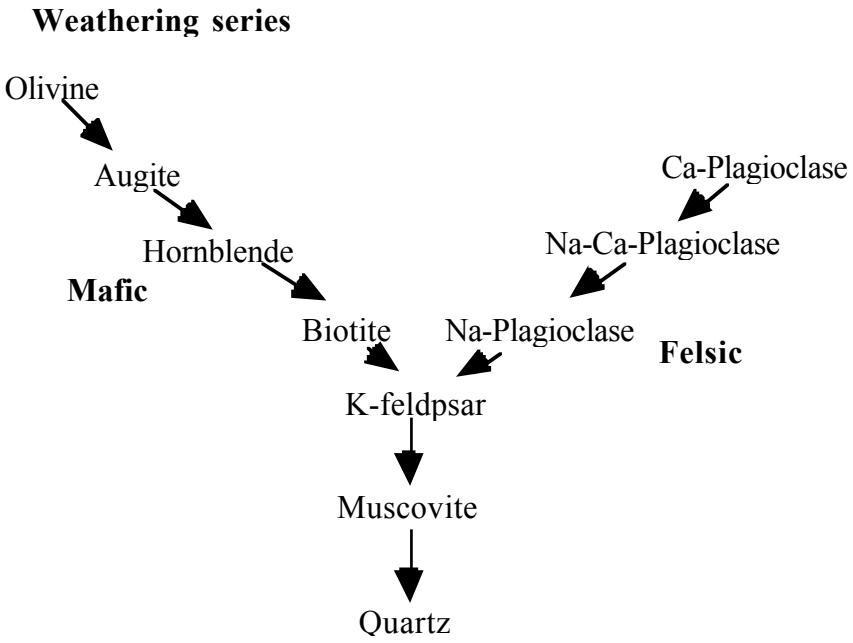
Determining the grain size range of unconsolidated sediment

Weathering process	Type of weathering product	Example	Ultimate depositional product
Physical weathering	Particulate residues	Silicate minerals such as quartz and feldspar; all types of rock fragments	Sandstones, conglomerates, mudrocks
Chemical weathering			
Hydrolysis	Soluble constituents	Silicic acid (H_4SiO_4); K^+, Na^+, Mg^{2+}, Ca^{2+}, etc.	Cherts, limestones, etc.
	Secondary minerals	Clay minerals	Mudrocks (shales)
Simple solution	Soluble constituents	Silicic acid; K^+, Na^+, Mg^{2+}, Ca^{2+}, HCO_3^-, SO_4^{2-}, etc.	Limestones, evaporites, chert, etc.
Oxidation	Secondary minerals	Ferric oxides (Fe_2OOH); manganese oxides (MnO_2)	Minor constituent in siliciclastic rocks
	Soluble constituents	Silicic acid; SO_4^{2-}	Chert, evaporites, etc.

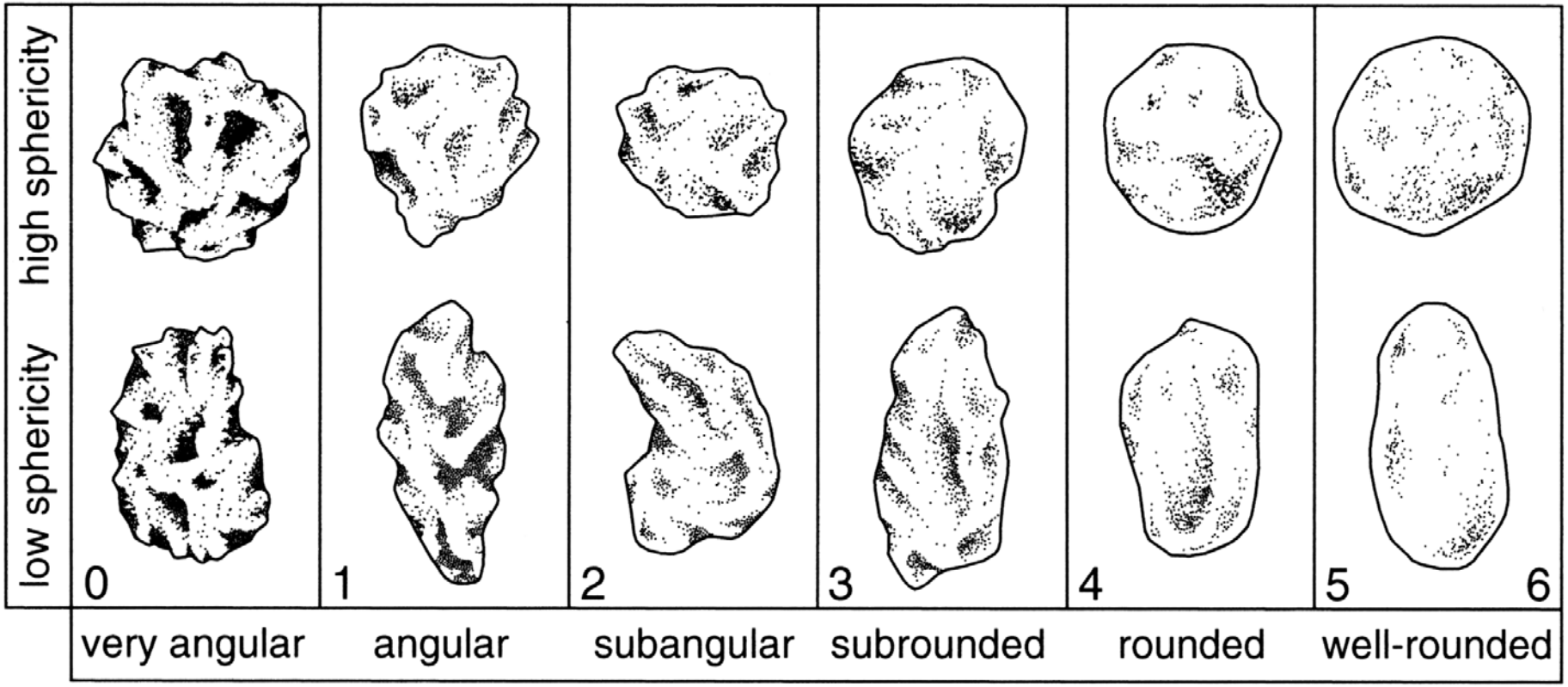
Principal kinds of subaerial weathering processes, their products, and the types of sedimentary rocks ultimately formed from these products



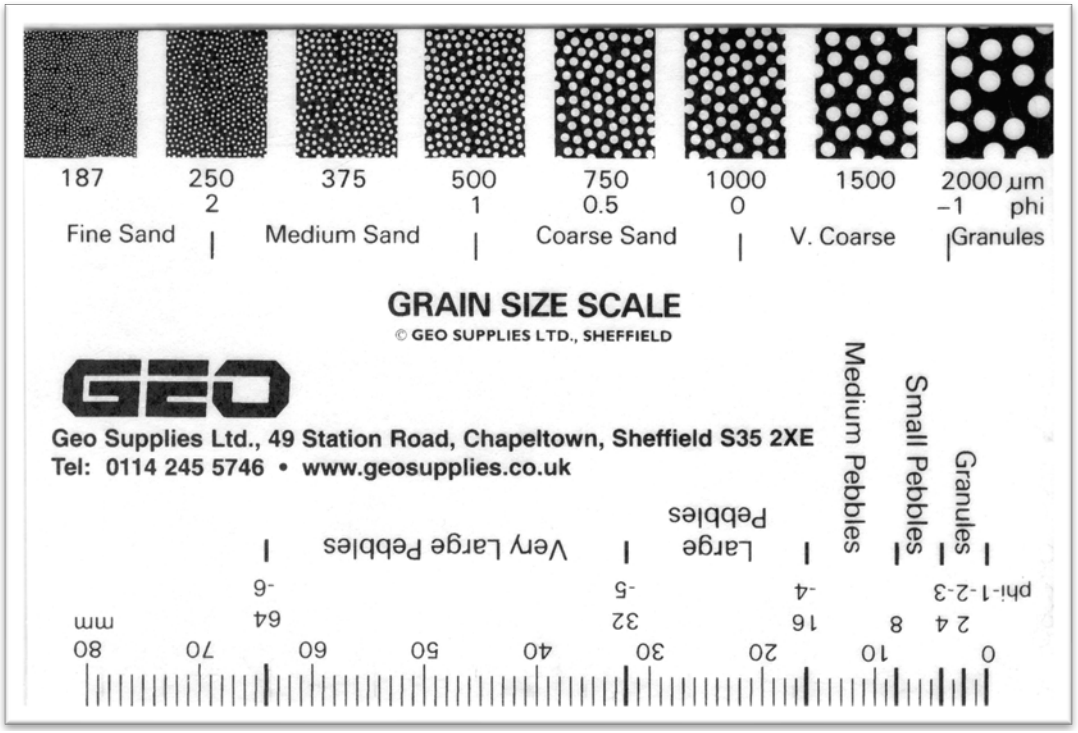
Mineralogical and chemical trends in a weathering profile of Mississippian age developed on a granodiorite near Boulder, Colorado, USA (after Wahlstrom, 1948)



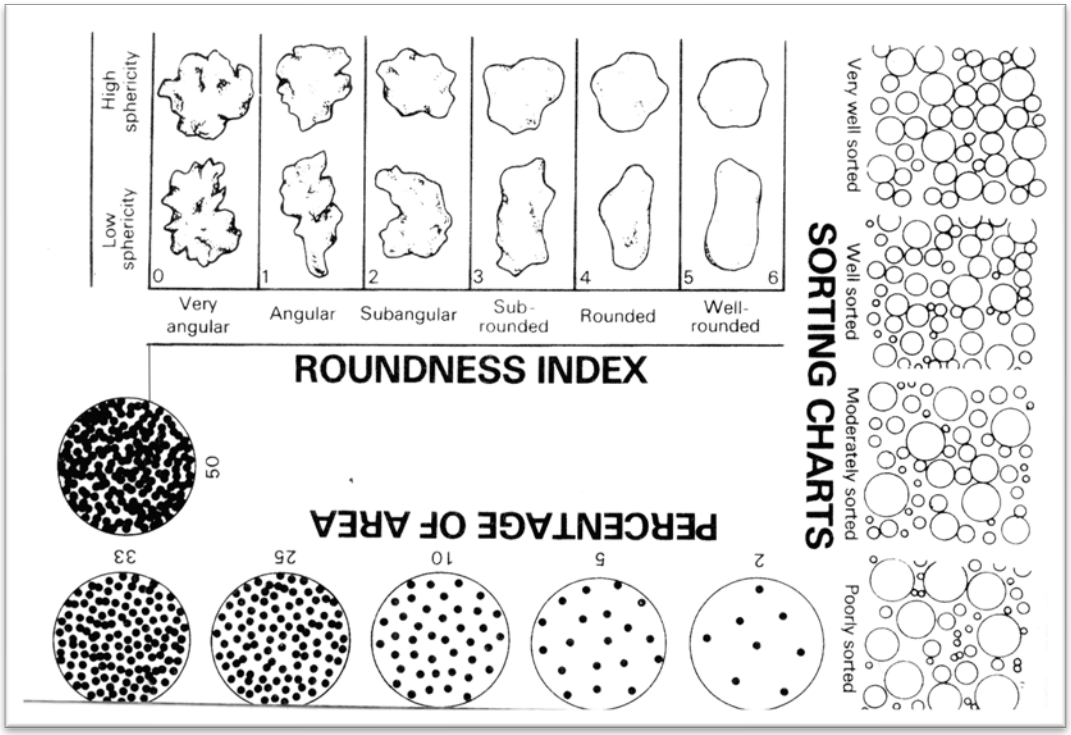
The 'weatherability' series for the common igneous silicate minerals. The series is the reverse of Bowen's reaction series for mineral crystallization from silicate melts (after Goldrich, 1938)



Categories of angularity for grains of low and high sphericity (after Pettijohn et al., 1973)



Front



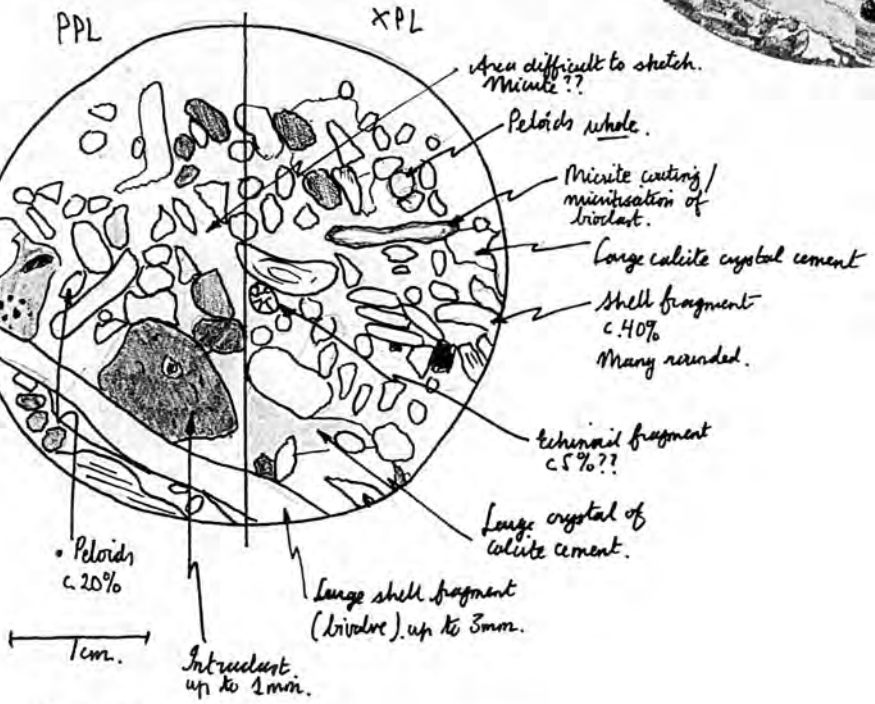
Back

Visual comparison aid used for determining the grain size and textural properties of sedimentary rocks. Use with a hand lens to determine the grain size.

Worked example: Thin section description



Thin section view



- GRAINS (>70%)
  - → Shell fragments 40% - fragmented, rounded, micritised, up to 3mm.
  - → Peloids 20-30% - whole, <1mm.
  - → Intraclasts - up to 2mm? 20%?
  - → Bryozoa + echinoids, fragmented, up to 1mm. 20%?
- MATRIX (<10%)
  - → Micrite between some grains only locally.
- CEMENT (c. 20%?)
  - → Large calcite crystals up to 2mm.

DESCRIPTIVE NAME : Bioclastic, peloidal packstone, with c 75% grains, 10% micrite matrix and c. 20% large crystals of calcite cement.

PRACTICAL 1

INTRODUCTION TO SILICICLASTIC SEDIMENTS AND ROCKS

LEARNING OUTCOMES

By the end of today's practical you should

- be able to use the Wentworth Scale to describe the grain size of sedimentary rocks in hand specimen
- be familiar with qualitative estimates for sediment sorting based on hand specimens
- understand the methods of analysis of grain size distributions of unconsolidated sediments and the parameters that are important in determining depositional processes
- confident in using triangular diagrams to classify sandstones
- able to recognize quartz and feldspar in hand specimens of sandstones
- be familiar with the appearance of a variety of finer grained sandstones, including wackes, micaceous sandstones and litharenites
- be able to recognize quartz and calcite cemented sandstones

A. GRAIN SIZE ANALYSIS

The grain size range of unconsolidated sediments can be analysed accurately by sieving. The table below gives the weight percentages of grains in each size class for two sediment samples.

Size **Sample 1**

Wentworth	mm	Ø	wt. %	cum. %
Pebble	16-8	-3	6.3	
	8-4	-2	11.3	
Granule	4-2	-1	20.1	
V coarse sand	2-1	0	32.1	
Coarse sand	1-0.5	+1	24.7	
Med sand	0.5-0.25	+2	4.4	
Fine sand	0.25-0.125	+3	0.8	
V fine sand	0.125-0.063	+4	0.3	
Mud	<0.063	+5		

Firstly, complete the cumulative % columns and then, using graph paper, plot a cumulative curve for Sample 1.

Note (a) coarsest grain sizes are plotted to the left
(b) the cumulative percentages are located at the smaller diameter for each grain size class.

Now read off from your graph the Ø values for the following percentiles. You will need these for your calculations.

Percentile **Sample 1**

Ø5
Ø16
Ø50
Ø84
Ø95

Average Grain Size

The most widely-used measure of average grain size is the 'mean diameter'. This is calculated by taking two or more percentiles and dividing by the number of percentiles used. The more percentiles used, the more meaningful the results. For thus sample use

$$\text{Mean diameter} = \frac{Ø16 + Ø50 + Ø84}{3}$$

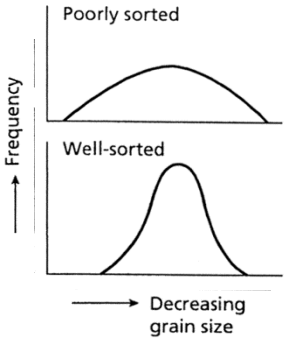
The mean grain size for sample 1 is .....Ø, which corresponds to

..... on the Wentworth descriptive scale

Sorting

The most commonly used measure of sorting is the **inclusive graphic standard deviation** (σ_1):

$$\sigma_1 = \frac{Ø84 - Ø16}{4} + \frac{Ø95 - Ø5}{6.6}$$



There are simpler methods, but they do not take such good account of the tails of the distribution.

Scale for sorting:

σ_1 <	0.35	very well sorted
	0.35 - 0.5	well sorted
	0.5 - 0.71	moderately well sorted
	0.71 - 1.0	moderately sorted
	1.0 - 2.0	poorly sorted
	2.0 - 4.0	very poorly sorted
>	4.0	extremely poorly sorted

Using the formula and scale above, work out the sorting of the sample, numerically and by name:

Sample 1 ....., which is therefore .....

Skewness

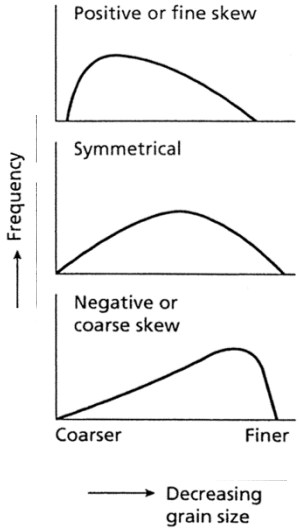
Skewness is a measure of the symmetry of the frequency curve. In symmetrical distributions the value will be close to zero, a negative value indicates a tail in the coarser grades and a positive value a tail in the finer grades. Values are always small, so a skewness of, say, ±0.2 is a significant departure from a normal distribution.

Calculate the skewness of the sample, using the formula:

$$\text{Skewness} = \frac{Ø16 + Ø84 - 2(Ø50)}{2(Ø84 - Ø16)} + \frac{Ø5 + Ø95 - 2(Ø50)}{2(Ø95 - Ø5)}$$

Sample 1 .....

Sample 1 has a tail in the ..... grades



B. SANDSTONE CLASSIFICATION: THEORY

Triangular diagrams are used in many areas of geology to show compositions of rocks and minerals. It is important that you understand how they are used in general, as well as the particular application to sandstone classification.

Seven sandstone analyses derived from point-counting thin sections under the microscope are tabulated below as percentages of the constituents. Start by deciding which of the triangles on the lecture handout is appropriate to each composition (in other words are they arenites, wackes or not sandstones at all?). Plot the position of each of the arenites on the triangular diagram on the next page. You should be able to determine the names of any rocks which are not arenites by using the lecture handout. Don't forget that you will have to re-calculate the raw data so that Quartz + Feldspar + Rock Fragments totals 100%.

Sample	Quartz	Feldspar	RF	Mica	Matrix	Cement	Pores
1	75	0	1	0	0	20	4
2	74	0	5	0	21	0	0
3	65	15	5	0	2	12	1
4	17	45	10	1	5	17	5
5	52	14	20	2	10	2	0
6	18	0	0	4	78	0	0
7	70	3	3	1	0	14	9

Which samples classify as arenites? ……, which as

wackes? …… and which as neither? ……

Now give each rock a name:

1…… 2 ……

3…… 4 ……

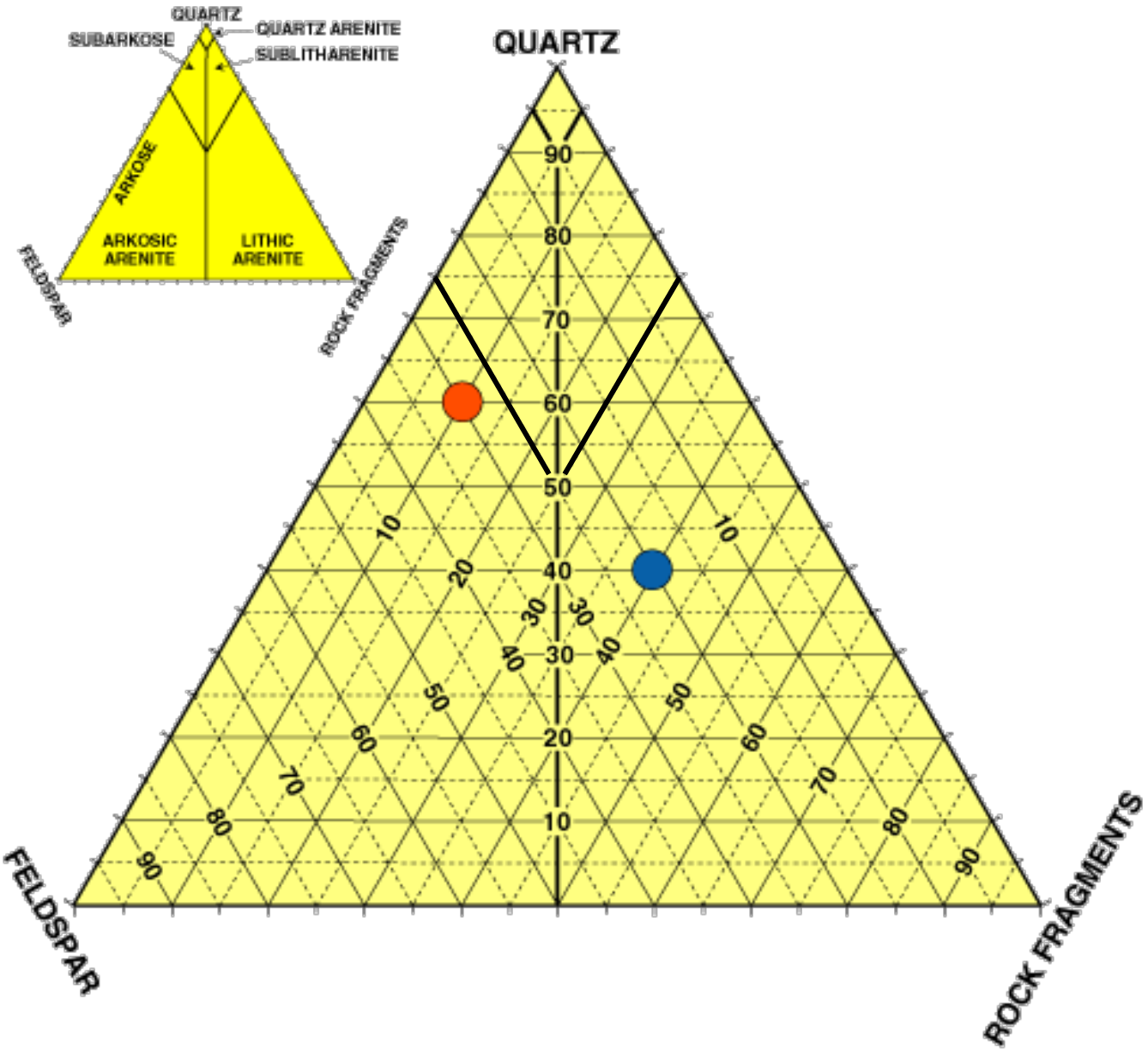
5…… 6 ……

7……

Arenite Classification
(for sandstones with <15% mud)

Examples of how to plot data:

- 60% Quartz
  30% Feldspar
  10% Rock fragments = ARKOSIC ARENITE
- 40% Quartz
  20% Feldspar
  40% Rock fragments = LITHIC ARENITE



C. APPLICATION TO SANDSTONE HAND SPECIMENS AND THIN SECTIONS

Feldspathic and quartz-rich sandstones

Look at the three sandstone hand specimens S4, R26 and D6. The average grain size of S4 is approximately …… mm which puts it in the ……Wentworth size class.

Examine the composition of S4 and R26 in hand specimen and thin section. S4 is almost entirely made of two constituents. What are they and what characteristics can you use to identify them in both hand specimen and thin section?

……

……

Estimate the percentage of each constituent using the comparison chart on the reverse side of your grain size card.

……

R26 has a similar composition although it is finer grained. Both these samples would probably classify as ……on the sandstone classification.

What other feature is shown by R26?

……

All three samples are red/purple because ……

D6 has an average grain size of …… mm, making it a …… (Wentworth Size Class).

What is the dominant mineral in this specimen? .....

There is also a small amount of feldspar present. What characteristics can you use to identify the feldspar in this sample?

……

Estimate the percentage of both the quartz and feldspar. What would be the most suitable name for this rock?

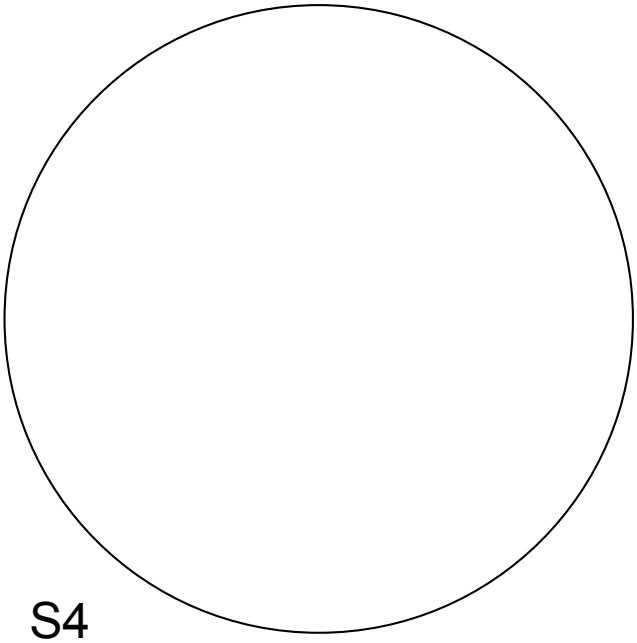
…….....

Comment on the roundness and sphericity of the quartz in D6 ……

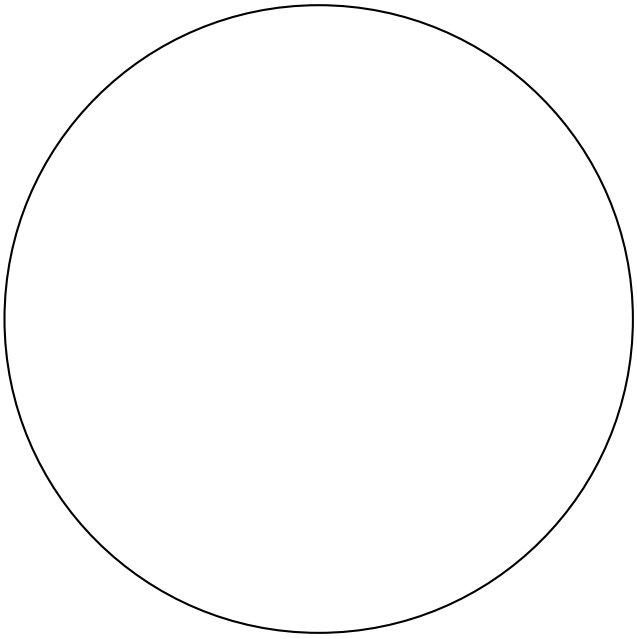
……

What does this suggest about transport mechanisms? ……

……



S4



R26

Sandstones with rock fragments and muddy sandstones

S20

Look carefully at this hand specimen with a lens and thin section with the microscope. The fresh broken surfaces are most revealing in hand specimen.

Firstly, estimate the grain size …… mm, making this a …… using the Wentworth class names. Now think about the composition. You should be able to recognize quartz and feldspar now (make sure you can!), but in S20 that leaves a lot of dark-coloured fragments. If these are not quartz or feldspar they must be rock fragments, although it is not possible to say of what sort of rock without recourse to a thin section. Now have a go at estimating the proportions of quartz, feldspar and rock fragments:

quartz ……. feldspar …… rock fragments ……..

This means that the rock would classify as a …… using the Dott triangles. Most samples of S20 have black streaks visible, probably along the bedding surfaces. What are these likely to be?

……

S10

At first sight S10 shows a few granules / pebbles set in a dark-coloured muddy matrix. With a lens you should be able to make out a proportion of sand-sized grains and the average grain size is probably in the sand range although this is a …… sorted rock. This rock is not likely

to be an arenite because ……..

and it would thus be called a …… . By what sort of

process would it have been deposited? ……

……

A(3) and S12

Examine these samples in both hand specimen and thin section.
On cut or freshly broken surfaces in hand specimen these two samples have a somewhat glassy appearance. This is characteristic of rocks where the detrital grains and cement are of the same mineral and it becomes difficult to pick out individual grains.

What mineral do you think is forming most of both grains and cement in

these two rocks? …… Both these rocks would

classify as…… Rocks of this type have commonly been called quartzites and the name still appears in many formation names, although the term is better restricted to metamorphosed quartz-rich sandstones.

In addition to the main constituent mineral, S12 contains a platy mineral prominent on some surfaces only. Identify the mineral and explain why it is

more obvious on some surfaces than others ……

……

Although the mineral is obvious on surfaces along which the rock has split, this mineral rarely forms more than a few percent of the rock and is not included in formal sandstone classifications. However, it is appropriate to

use the adjective …… in front of the rock name.

S17

What sedimentary structure is present in this sample? ……

Assuming that the coarser material defines the base of the sample, S17

grades from …….. at the base to ……..

at the top (use Wentworth size names). What is the composition of this rock and what name would therefore be appropriate?

……

S14

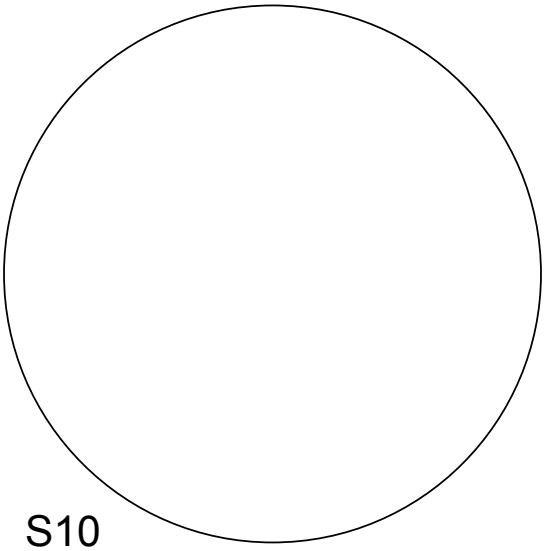
At first sight this sample is not very revealing. Try a drop of acid on it.

What mineral is present? …… Now look at the weathered surface with a lens. You should be able to see lots of small (fine sand sized?) quartz grains. If you don't believe quartz is present test the

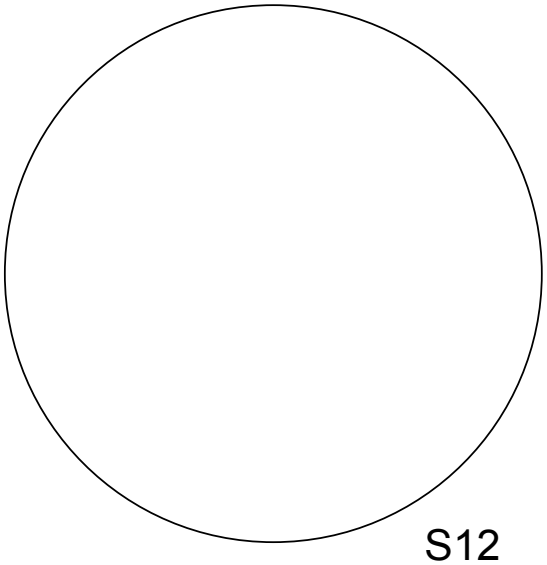
sample for hardness. Confirm your identification in thin section.

If the grains are quartz, the calcite is a ……,

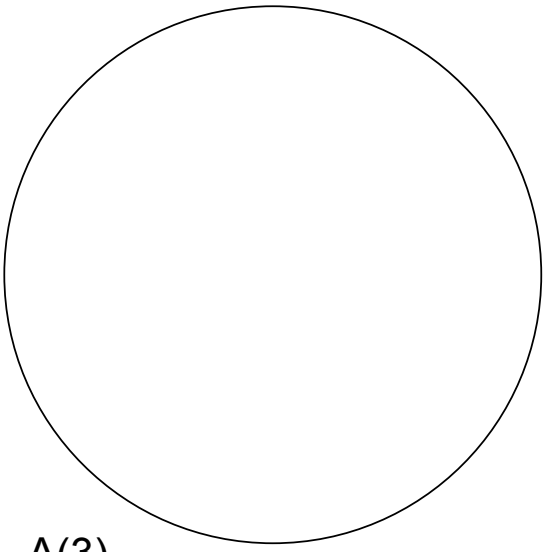
so a suitable name for this rock would be……



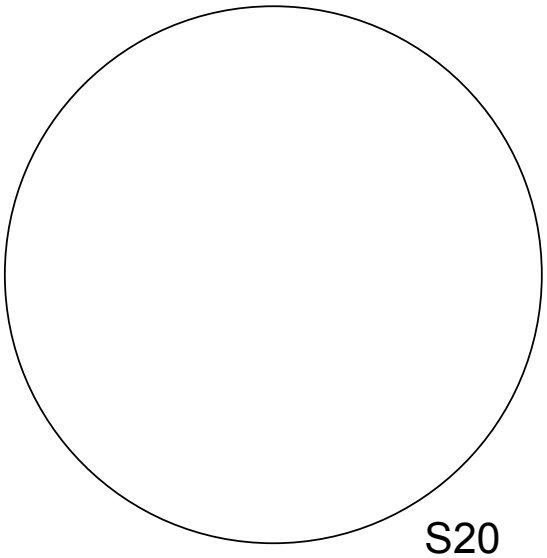
S10



S12



A(3)



S20